

Assignment-02

1. Consider the function $f(x, y, z) = \tan^{-1}\left(\frac{z}{x^2y^2}\right)$ to estimate the slopes in the direction of coordinate axes at the point $(1, 1, -1)$. Discuss the increasing and decreasing of the given function according to the slopes that you estimated.
2. Assume the function $f(x, y, z) = e^{-xy} \ln\left(\frac{1}{z}\right)$ to find rate of changes in the direction of coordinate axes at the point $(-1, 0, 2)$.
3. Show that $f_{xyz} = f_{yzx} = f_{zxy}$ for $f(x, y, z) = \ln(x^2y^3z^4)$.
4. Find the rate of increase/decrease of the function $z = \sqrt{x^2 + y^2}$ in the direction of coordinate axes at the point $(3, -4)$.
5. Show that $u(x, t) = \alpha \sin(kx - mt)$ is a solution of the one dimensional wave equation $u_{tt} = \frac{m^2}{k^2} u_{xx}$.
6. Find the value of ω for which $u(x, t) = e^{-2t} \sin\left(\frac{x}{2}\right)$ is a solution of the one dimensional heat equation $u_t = \omega u_{xx}$.
7. Determine whether $u = e^{-x}(x \sin y - y \cos y)$ and $v = e^{-x}(y \sin y + x \cos y)$ satisfy the Cauchy-Riemann equations $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$.
8. For $x^3 + y^3 = 3axy$ find $\frac{dy}{dx}$ and verify your result by the implicit differentiation.
9. For $\ln(x^2 + y^2) + 2z = \frac{1}{e}$ find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$. Also, verify your result through the implicit differentiation.
10. Show that $\left(\frac{\partial w}{\partial x}\right)^2 + \left(\frac{\partial w}{\partial y}\right)^2 + \left(\frac{\partial w}{\partial z}\right)^2 = \left(\frac{x^2y^2 + y^2z^2 + z^2x^2}{4xyz}\right) \left(\frac{dw}{dr}\right)^2$, it is given that $w = f(r)$ is a differentiable function with $r = \sqrt{xyz}$.
11. For $\omega = f(x^2 - 2y + 4z)$ show that $\frac{1}{x} \frac{\partial \omega}{\partial x} = \frac{\partial \omega}{\partial y} + \frac{\partial \omega}{\partial z}$.
12. Suppose $z = e^{xy}$, where $x = \sin t$ and $y = \cos t$. Find the rate of change of z along the two-dimensional coordinate axes at the point $t = \frac{\pi}{4}$.